MATHS / ALGEBRA / FACTORISING

Two-sided classroom worksheet

PAGE 1 · HIGHEST COMMON FACTOR

EVENS: CLASS / ODDS: HOME

Numerical method: divide every term by the same HCF.

HCF — two algebraic terms

Example: 12x + 18 = 6(2x + 3)

1. $8x + 12$
2. $12x^2 + 18x$
3. $14xy - 21x$
4. $18x^3y^2 + 24x^2y$
5. $20a^2b + 30ab^2$
6. $24x^4y^3 - 36x^2y^2$
7. $35m^3n + 49m^2n^2$
8. $30x^5y^3z - 45x^4y^2z^2$

HCF — three algebraic terms

Example: 6x + 12y + 18 = 6(x + 2y + 3)

1. $6x + 12y + 18$
2. $12x^2 + 18x + 30x^3$
3. $15xy + 20xz - 25x$
4. $18x^3y^2 + 24x^2y^3 - 30x^2y^2$
5. $21p^2q + 28pq^2 + 35pq$
6. $24a^3b^2 - 36a^2b^3 + 48a^2b^2$
7. $16x^5y^2 + 24x^3y^3 - 40x^4y^2$
8. $-30x^5y^3z + 45x^4y^2z^2 + 15x^3y^4z$

Check by multiplying out the brackets.

A negative factor reverses both signs.

PAGE 2 · FACTORISING BY GROUPING

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Make the two brackets identical; sometimes take out a negative.

Grouping — matching brackets

Example: $xy + 2y + xz + 2z = (x + 2)(y + z)$

1. $xy + xz + 2y + 2z$
2. $2x^2 + 6xy + 3xz + 9yz$
3. $ax + ay + bx + by$
4. $6x^2 + 9xy + 4xz + 6yz$
5. $x^3 + 2x^2 + 3x + 6$
6. $4x^4y + 8x^3y^2 + 3x^2z + 6xyz$
7. $a^2b + 3ab^2 + 2ac + 6bc$
8. $10x^5y^2 + 15x^4y^3 + 6x^2z + 9xyz$

Grouping — take out a negative

Example: $x(y-z) + 3(z-y) = (x-3)(y-z)$

1. $xy - xz + 3z - 3y$
2. $x^2 - xy - 5x + 5y$
3. $a^2 + 2ab - 3ac - 6bc$
4. $4x^3 - 8x^2y - 3xz + 6yz$
5. $2ab - 4ac + 6c - 3b$
6. $6x^4y^2 - 9x^3y^3 - 4xz + 6yz$
7. $x^5 - x^4y - 7x + 7y$
8. $12a^5b^2 - 18a^4b^3 - 20a^2c + 30abc$

Check by multiplying out the brackets.

A negative factor reverses both signs.